

Table 1: Sample statements and questions from tasks 1 to 10.

Task 1: Single Supporting Fact Mary went to the bathroom. John moved to the hallway. Mary travelled to the office. Where is Mary? A:office	Task 2: Two Supporting Facts John is in the playground. John picked up the football. Bob went to the kitchen. Where is the football? A:playground
Task 3: Three Supporting Facts John picked up the apple. John went to the office. John went to the kitchen. John dropped the apple. Where was the apple before the kitchen? A:office	Task 4: Two Argument Relations The office is north of the bedroom. The bedroom is north of the bathroom. The kitchen is west of the garden. What is north of the bedroom? A: office What is the bedroom north of? A: bathroom
Task 5: Three Argument Relations Mary gave the cake to Fred. Fred gave the cake to Bill. Jeff was given the milk by Bill. Who gave the cake to Fred? A: Mary Who did Fred give the cake to? A: Bill	Task 6: Yes/No Questions John moved to the playground. Daniel went to the bathroom. John went back to the hallway. Is John in the playground? A:no Is Daniel in the bathroom? A:yes
Task 7: Counting Daniel picked up the football. Daniel dropped the football. Daniel got the milk. Daniel took the apple. How many objects is Daniel holding? A: two	Task 8: Lists/Sets Daniel picks up the football. Daniel drops the newspaper. Daniel picks up the milk. John took the apple. What is Daniel holding? milk, football
Task 9: Simple Negation Sandra travelled to the office. Fred is no longer in the office. Is Fred in the office? A:no Is Sandra in the office? A:yes	Task 10: Indefinite Knowledge John is either in the classroom or the playground. Sandra is in the garden. Is John in the classroom? A:maybe Is John in the office? A:no

Simple Negation and Indefinite Knowledge Tasks 9 and 10 test slightly more complex natural language constructs. Task 9 tests one of the simplest forms of negation, that of supporting facts that imply a statement is false e.g. “*Fred is no longer in the office*” rather than “*Fred travelled to the office*”. (In this case, task 6 (yes/no questions) is a prerequisite to the task.) Task 10 tests if we can model statements that describe possibilities rather than certainties, e.g. “*John is either in the classroom or the playground.*”, where in that case the answer is “*maybe*” to the question “*Is John in the classroom?*”.

Basic Coreference, Conjunctions and Compound Coreference Task 11 tests the simplest type of coreference, that of detecting the nearest referent, e.g. “*Daniel was in the kitchen. Then he went to the studio.*”. Real-world data typically addresses this as a labeling problem and studies more sophisticated phenomena (Soon et al., 2001), whereas we evaluate it as in all our other tasks as a question answering problem. Task 12 (conjunctions) tests referring to multiple subjects in a single statement, e.g. “*Mary and Jeff went to the kitchen.*”. Task 13 tests coreference in the case where the pronoun can refer to multiple actors, e.g. “*Daniel and Sandra journeyed to the office. Then they went to the garden.*”.

Time Reasoning While our tasks so far have included time implicitly in the *order* of the statements, task 14 tests understanding the use of time expressions within the statements, e.g. “*In the afternoon Julie went to the park. Yesterday Julie was at school.*”, followed by questions about the order of events such as “*Where was Julie before the park?*”. Real-world datasets address the task of evaluating time expressions typically as a labeling, rather than a QA task, see e.g. UzZaman et al. (2012).

Basic Deduction and Induction Task 15 tests basic deduction via inheritance of properties, e.g. “*Sheep are afraid of wolves. Gertrude is a sheep. What is Gertrude afraid of?*”. Task 16 similarly

Table 2: Sample statements and questions from tasks 11 to 20.

Task 11: Basic Coreference Daniel was in the kitchen. Then he went to the studio. Sandra was in the office. Where is Daniel? A:studio	Task 12: Conjunction Mary and Jeff went to the kitchen. Then Jeff went to the park. Where is Mary? A: kitchen Where is Jeff? A: park
Task 13: Compound Coreference Daniel and Sandra journeyed to the office. Then they went to the garden. Sandra and John travelled to the kitchen. After that they moved to the hallway. Where is Daniel? A: garden	Task 14: Time Reasoning In the afternoon Julie went to the park. Yesterday Julie was at school. Julie went to the cinema this evening. Where did Julie go after the park? A:cinema Where was Julie before the park? A:school
Task 15: Basic Deduction Sheep are afraid of wolves. Cats are afraid of dogs. Mice are afraid of cats. Gertrude is a sheep. What is Gertrude afraid of? A:wolves	Task 16: Basic Induction Lily is a swan. Lily is white. Bernhard is green. Greg is a swan. What color is Greg? A:white
Task 17: Positional Reasoning The triangle is to the right of the blue square. The red square is on top of the blue square. The red sphere is to the right of the blue square. Is the red sphere to the right of the blue square? A:yes Is the red square to the left of the triangle? A:yes	Task 18: Size Reasoning The football fits in the suitcase. The suitcase fits in the cupboard. The box is smaller than the football. Will the box fit in the suitcase? A:yes Will the cupboard fit in the box? A:no
Task 19: Path Finding The kitchen is north of the hallway. The bathroom is west of the bedroom. The den is east of the hallway. The office is south of the bedroom. How do you go from den to kitchen? A: west, north How do you go from office to bathroom? A: north, west	Task 20: Agent’s Motivations John is hungry. John goes to the kitchen. John grabbed the apple there. Daniel is hungry. Where does Daniel go? A:kitchen Why did John go to the kitchen? A:hungry

tests basic induction via inheritance of properties. A full analysis of induction and deduction is clearly beyond the scope of this work, and future tasks should analyse further, deeper aspects.

Positional and Size Reasoning Task 17 tests spatial reasoning, one of many components of the classical SHRDLU system (Winograd, 1972) by asking questions about the relative positions of colored blocks. Task 18 requires reasoning about the relative size of objects and is inspired by the commonsense reasoning examples in the Winograd schema challenge (Levesque et al., 2011).

Path Finding The goal of task 19 is to find the path between locations: given the description of various locations, it asks: how do you get from one to another? This is related to the work of Chen & Mooney (2011) and effectively involves a search problem.

Agent’s Motivations Finally, task 20 questions, in the simplest way possible, *why* an agent performs an action. It addresses the case of actors being in a given state (hungry, thirsty, tired, ...) and the actions they then take, e.g. it should learn that hungry people might go to the kitchen, and so on.

As already stated, these tasks are meant to foster the development and understanding of machine learning algorithms. A single model should be evaluated across all the tasks (not tuning per task) and then the same model should be tested on additional real-world tasks.

In our data release, in addition to providing the above 20 tasks in English, we also provide them (i) in Hindi; and (ii) with shuffled English words so they are no longer readable by humans. A good learning algorithm should perform similarly on all three, which would likely not be the case for a method using external resources, a setting intended to mimic a learner being first presented a language and having to learn from scratch.

4 SIMULATION

All our tasks are generated with a simulation which behaves like a classic text adventure game. The idea is that generating text within this simulation allows us to ground the language used into a coherent and controlled (artificial) world. Our simulation follows those of Bordes et al. (2010); Weston et al. (2014) but is somewhat more complex.

The simulated world is composed of entities of various types (locations, objects, persons. etc.) and of various actions that operate on these entities. Entities have internal states: their location, whether they carry objects on top or inside them (e.g., tables and boxes), the mental state of actors (e.g. hungry), as well as properties such as size, color, and edibility. For locations, the nearby places that are connected (e.g. what lies to the east, or above) are encoded. For actors, a set of pre-specified rules per actor can also be specified to control their behavior, e.g. if they are hungry they may try to find food. Random valid actions can also be executed if no rule is set, e.g. walking around randomly.

The actions an actor can execute in the simulation consist of the following: *go* <location>, *get* <object>, *get* <object1> *from* <object2>, *put* <object1> *in/on* <object2>, *give* <object> *to* <actor>, *drop* <object>, *set* <entity> <state>, *look*, *inventory* and *examine* <object>. A set of universal constraints is imposed on those actions to enforce coherence in the simulation. For example an actor cannot get something that they or someone else already has, they cannot go to a place that is not connected to the current location, cannot drop something they do not already have, and so on. Using the underlying actions, rules for actors, and their constraints, defines how actors act. For each task we limit the actions needed for that task, e.g. task 1 only needs *go* whereas task 2 uses *go*, *get* and *drop*. If we write the commands down this gives us a very simple “story” which is executable by the simulation, e.g., *joe go playground; bob go office; joe get football*. This example corresponds to task 2. The system can then ask questions about the state of the simulation e.g., *where john?*, *where football?* and so on. It is easy to calculate the true answers for these questions as we have access to the underlying world.

To produce more natural looking text with lexical variety from statements and questions we employ a simple automated grammar. Each verb is assigned a set of synonyms, e.g., the simulation command *get* is replaced with either *picked up*, *got*, *grabbed* or *took*, and *drop* is replaced with either *dropped*, *left*, *discarded* or *put down*. Similarly, each object and actor can have a set of replacement synonyms as well, e.g. replacing Daniel with *he* in task 11. Adverbs are crucial for some tasks such as the time reasoning task 14.

There are a great many aspects of language not yet modeled. For example, all sentences are so far relatively short and contain little nesting. Further, the entities and the vocabulary size is small (150 words, and typically 4 actors, 6 locations and 3 objects used per task). The hope is that defining a set of well defined tasks will help evaluate models in a controlled way within the simulated environment, which is hard to do with real data. That is, these tasks are not a substitute for real data, but should complement them, especially when developing and analysing algorithms.

5 EXPERIMENTS

We compared the following methods on our tasks (on the English dataset): (i) an N -gram classifier baseline, (ii) LSTMs (long short term memory Recurrent Neural Networks) (Hochreiter & Schmidhuber, 1997), (iii) Memory Networks (MemNNs) (Weston et al., 2014), (iv) some extensions of Memory Networks we will detail; and (v) a structured SVM that incorporates external labeled data from existing NLP tasks. These models belong to three separate tracks. Weakly supervised models are only given question answer pairs at training time, whereas strong supervision provides the set of supporting facts at training time (but not testing time) as well. Strongly supervised ones give accuracy upper bounds for weakly supervised models, i.e. the performance should be superior given the same model class. Methods in the last external resources track can use labeled data from other sources rather than just the training set provided, e.g. coreference and semantic role labeling tasks, as well as strong supervision. For each task we use 1000 questions for training, and 1000 for testing, and report the test accuracy. We consider a task successfully passed if $\geq 95\%$ accuracy is obtained³.

³The choice of 95% (and 1000 training examples) is arbitrary.